

GRADUATE THESIS DEFENSE

SFDCT: A Frequency-Aware Deepfake Detector for Banking

Two-Stream Architecture with Block-DCT Frequency Branch and Gated Cross-Attention Fusion

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Computer Science & Artificial Intelligence

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THESIS

Defense Overview

01 Project Overview
Problem, Motivation, Objectives & Scope

02 Methodology
Frequency Hypothesis, Block-DCT, Gated Fusion, SFDCT-HFF & Liveness

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04 Conclusion
Contributions, Limitations & Future Directions



eKYC Deepfake Defense

01

CHAPTER ONE

Project Overview

Problem, Motivation, Objectives & Scope

1.1 Problem and Motivation

1.2 Objectives and Scope

Problem and Motivation: The Generalisation Gap

1 eKYC: Face Verification as the First Lock

eKYC opens accounts and confirms payments from one portrait photo. Face verification is the lock on that first door to financial services.

2 Two Distinct Threats

Deepfake (GAN face swap/synthesis) and **Presentation Attack** (print/replay). Either bypass means identity fraud and direct financial loss.

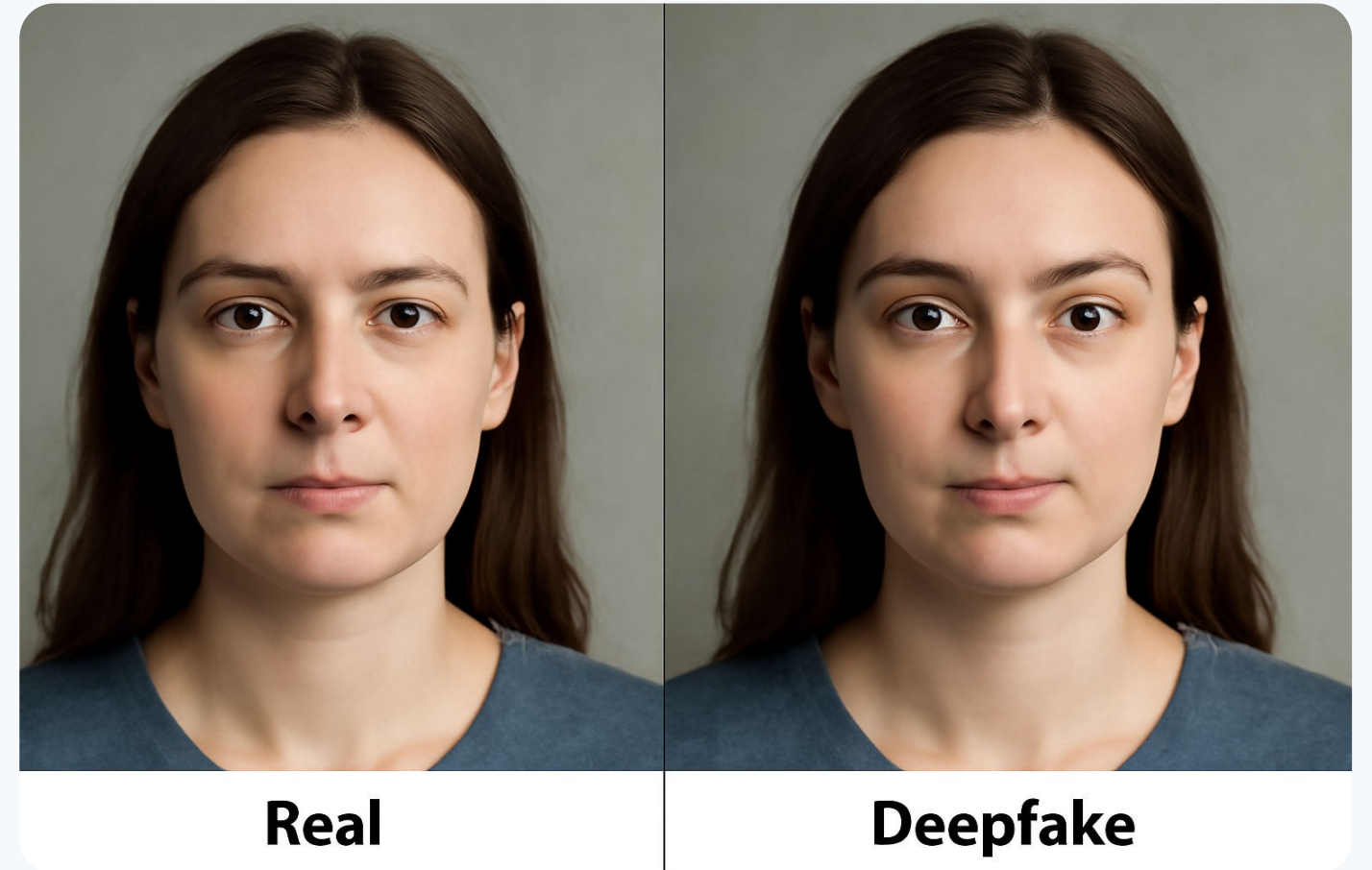
3 Regulatory Context: Circular 17/2024/TT-NHNN

Mandates biometric checks but fixes no error rate. The operational threshold is left to the system designer.

4 Core Difficulty: The Generalisation Gap

Pixel detectors collapse on unseen forgeries. Attackers always use newer tools than those in the training set.

! Key message: Face verification is the lock on the first door to financial services. The core problem is not detecting known forgeries but generalising to unseen ones — attackers always use newer tools.



Real vs. Deepfake face — nearly identical to the human eye

Objectives and Scope: Two-Layer Protection System

Four Objectives

Obj 1 — Web Platform

Simple platform to upload or capture a face, fast result

Obj 2 — Deepfake Detector (Main Contribution)

Fusing a block-DCT frequency branch with a CNN backbone via a gated attention, returning a heat map

Obj 3 — Liveness Utility

Secondary: live person vs. printed photo or screen replay

Obj 4 — Cloud Deployment

AWS EC2 deployment for always-on service

Scope & Boundaries

In Scope

- Frame-level aligned faces
- No temporal information
- Cross-dataset: FF++ to Celeb-DF-v2

Out of Scope

- Video-level temporal modelling
- 3D physical mask attacks
- Audio deepfake detection
- Direct injection attacks

Evaluation Protocol

DeepfakeBench: train FF++, test Celeb-DF-v2

Key message: The main contribution is the block-DCT frequency branch fused via gated attention. The system is evaluated honestly by cross-dataset protocol: train on FF++, test on Celeb-DF-v2.

02

CHAPTER TWO

Methodology

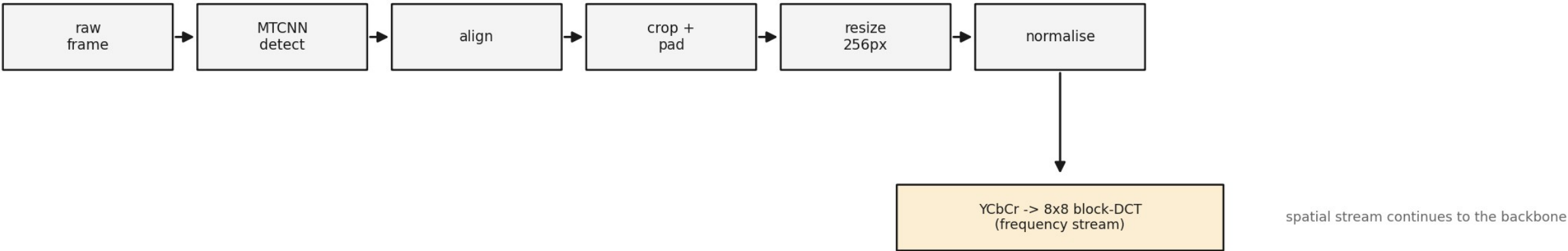
Frequency Hypothesis, Block-DCT, Gated Fusion, SFDCT-HFF & Liveness

2.1 Two-Stream SFDCT Architecture

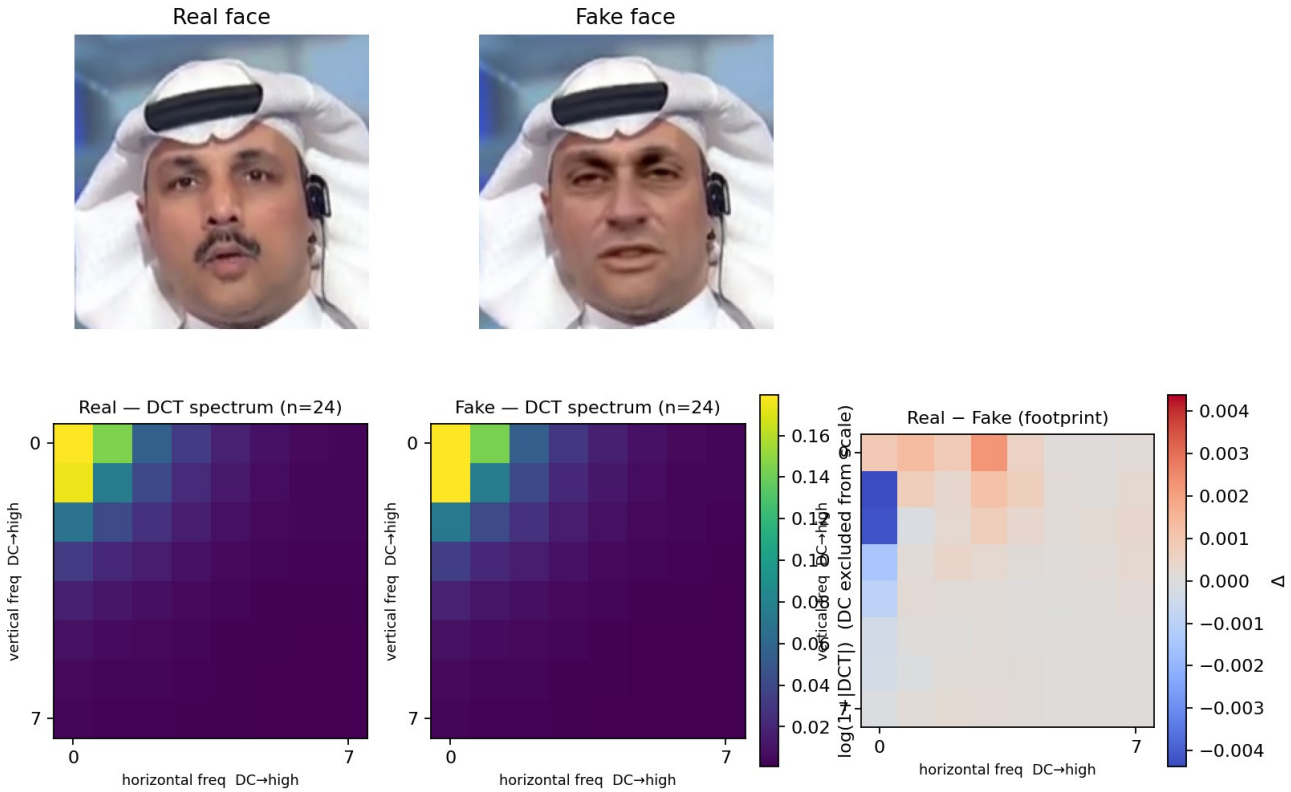
2.2 Block-DCT Descriptor

2.3 Gated Cross-Attention Fusion

2.4 SFDCT-HFF & Liveness

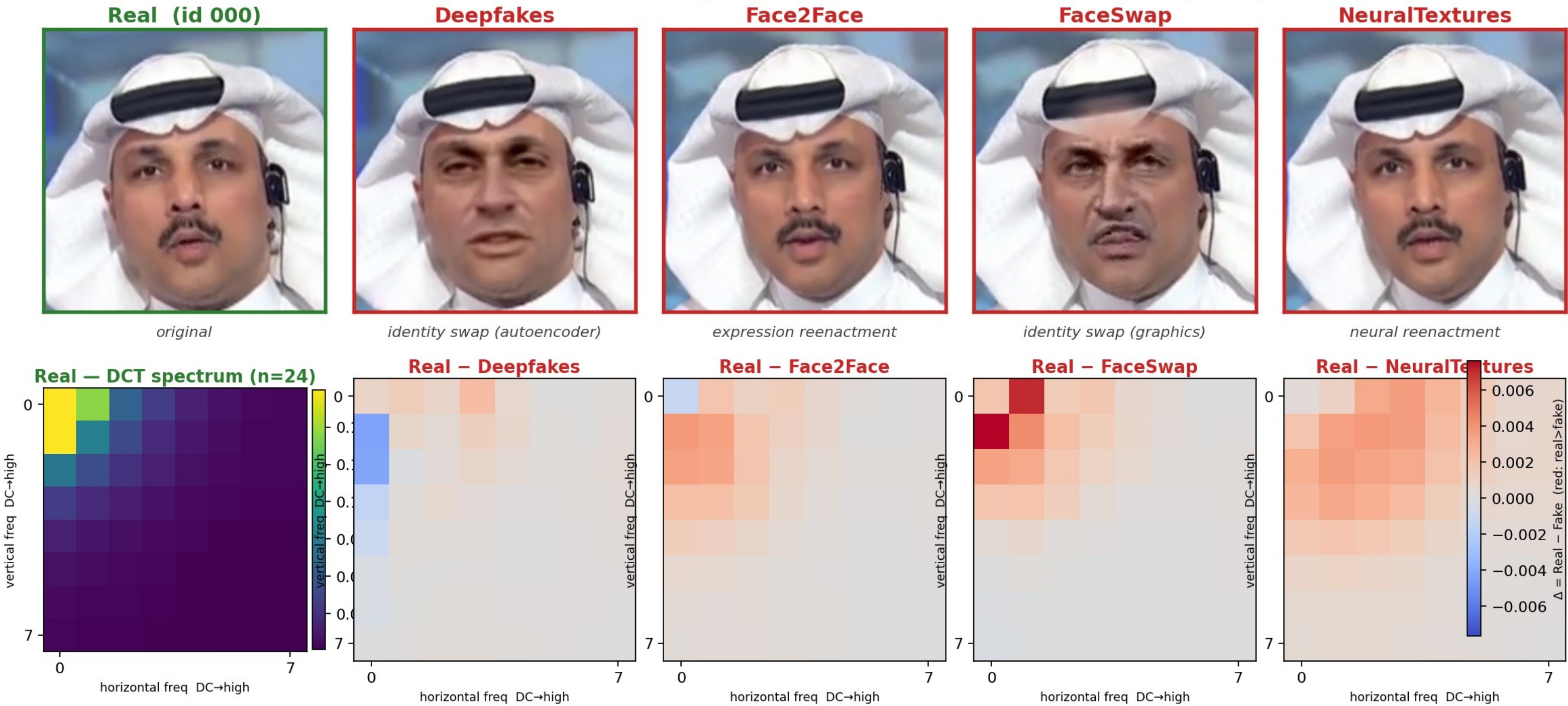


Real vs fake — block-DCT 8×8 spectrum (FF++ Deepfakes, id 000)

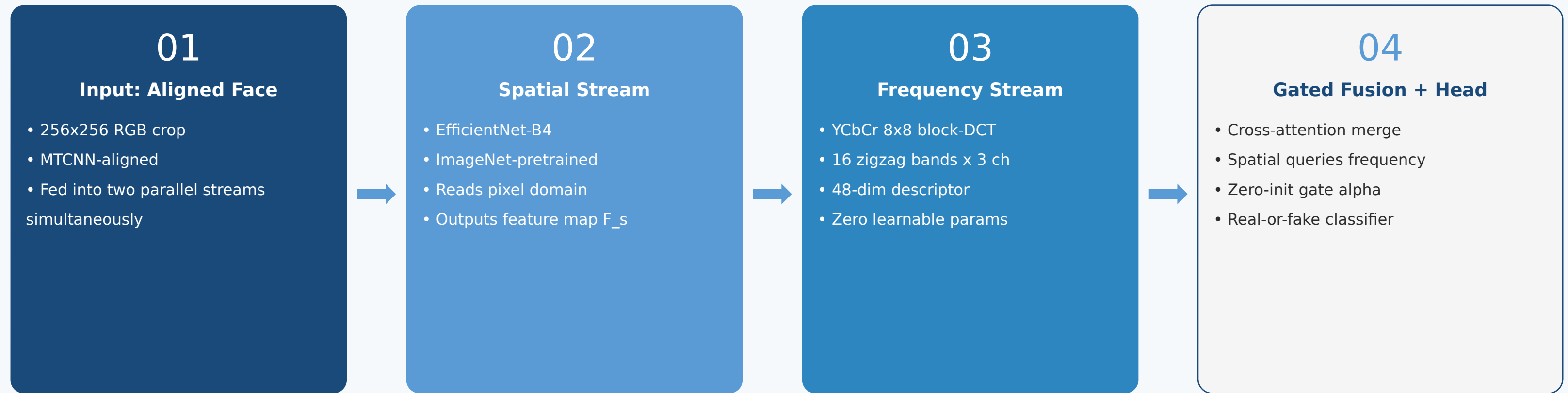


Forgery Footprints — FF++ Manipulations

FaceForensics++ (id 000) after preprocessing — faces + block-DCT footprint per method



The Frequency Hypothesis and Two-Stream SFDCT

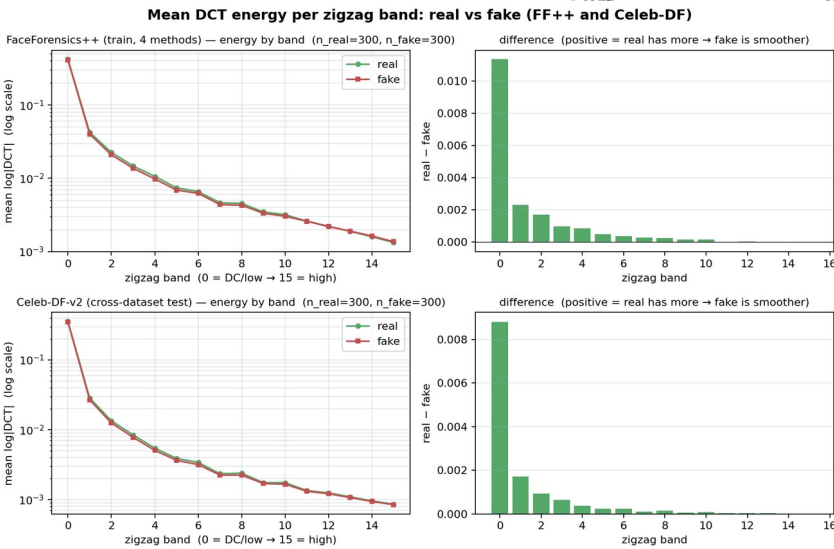
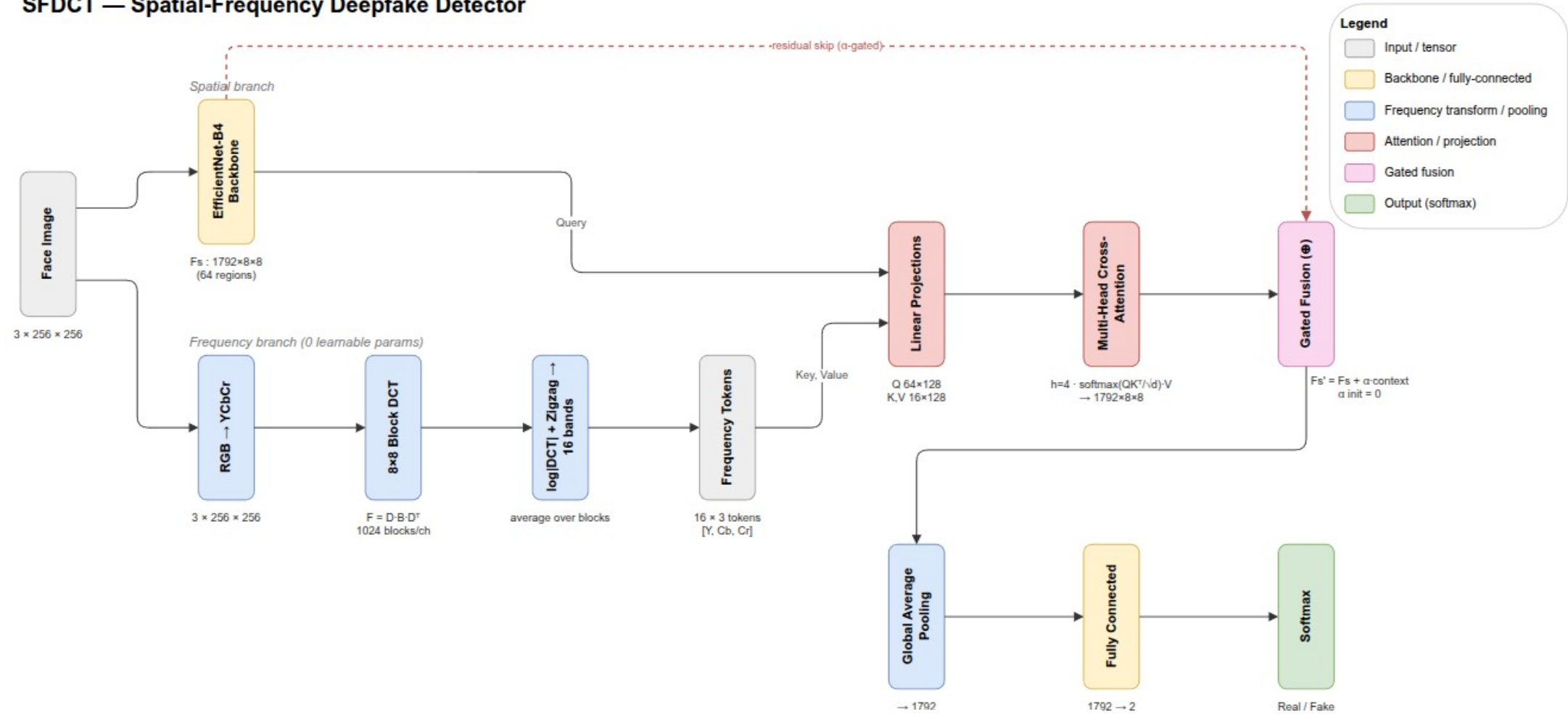


The Frequency Hypothesis

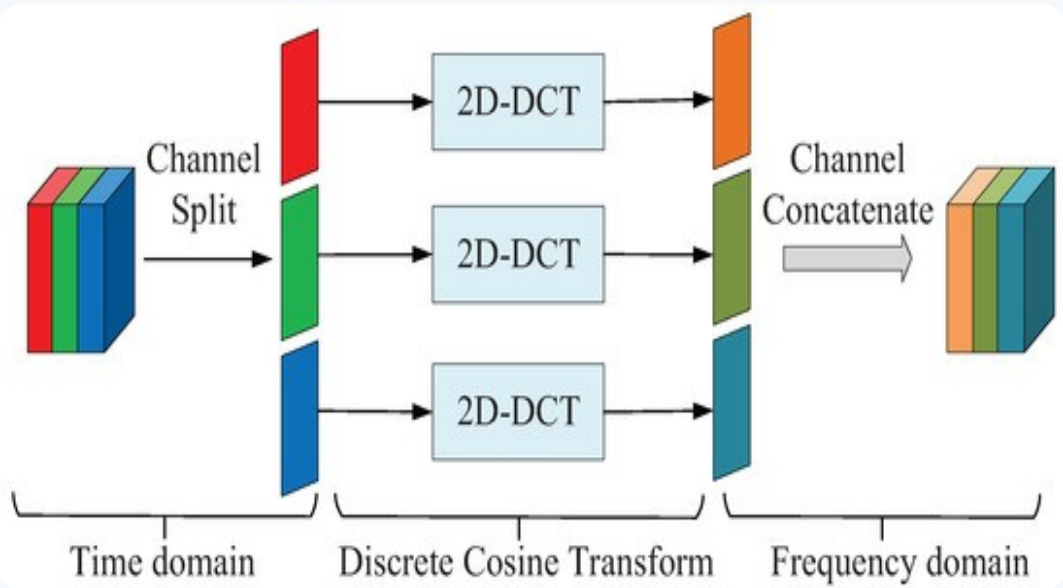
- **Blending boundary:** face-swap paste leaves a seam detectable in mid-frequency bands
- **Upsampling artifact:** GAN generators leave periodic patterns in high-frequency bands
- **Frequency inconsistency:** spliced sources break the unified frequency signature of real camera capture

Key message: Forgery traces are faint in pixels but loud in mid/high DCT bands. SFDCT reads both views — spatial (pixels) and frequency (DCT) — then merges them via a learned gate that starts closed.

SFDCT — Spatial-Frequency Deepfake Detector



From 8x8 Block-DCT to a 48-Dim Descriptor



Channel-wise 2D-DCT: Time domain to Frequency domain

Descriptor Construction Pipeline

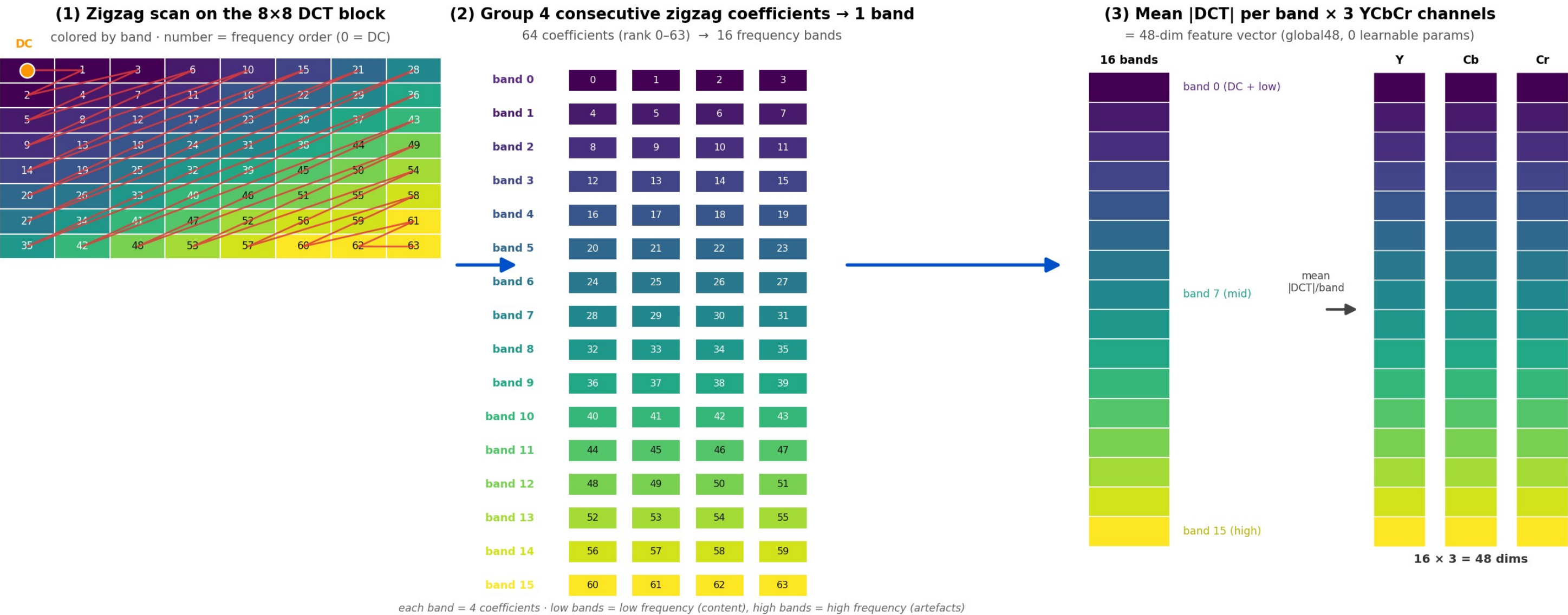
- 1. **Colour space conversion** — RGB to YCbCr to separate luminance from chrominance
- 2. **Block partition** — Non-overlapping 8x8 blocks (matches JPEG grid)
- 3. **Per-block 2D-DCT** — 64 coefficients per block
- 4. **Log-magnitude rescale** — $\log(1+|\text{coeff}|)$ prevents DC from swamping artefacts
- 5. **Zigzag grouping** — 64 coeffs grouped into 16 bands ($\text{floor}(\text{rank}/4)$)
- 6. **Per-band mean** — 16 bands x 3 channels = **48-dim descriptor**

Band Range	Frequency	Content	Forgery Trace	Action
0 (DC only)	DC component	Mean intensity	None	Dropped
1 - 3	Low	Coarse structure, identity	Minimal	Kept (SFDCT) / Dropped (HFF)
4 - 15	Mid - High	Texture, edges	Blending, upsampling, compression	Kept

Key message: The 48-dim descriptor is fixed (zero learnable parameters), cheap to compute, stable across datasets, and interpretable — each dimension maps to a known frequency band.

Block-DCT Descriptor — Zigzag to 48-D

From zigzag scan to 16 frequency bands and the 48-dim feature vector (ContentDCT)



ContentDCT drops DC + a few lowest bands (drop_low_bands) to remove content/identity leakage, emphasising the mid-high bands where forgery traces live.

Gated Cross-Attention: Space Queries Frequency

Cross-Attention Mechanism (Eq. 2.4)

- **Query (Q)** = Spatial feature map F_s from EfficientNet-B4
- **Key (K)** = Block-DCT descriptor D
- **Value (V)** = Block-DCT descriptor D
- Each spatial position retrieves a **weighted frequency summary**
- Not a flat concatenation — selective, position-aware fusion

Gated Residual (Eq. 2.5)

$$F_{\text{fused}} = F_s + \alpha \times \text{context}$$

- Scalar gate **alpha initialised to 0**
- At step 1: fusion = backbone exactly (floor guarantee)
- Gate opens **only where** frequency context lowers loss
- If frequency is noisy: gradient pulls alpha back to ~ 0

Learned Gate Statistics (after training)

1792

channels

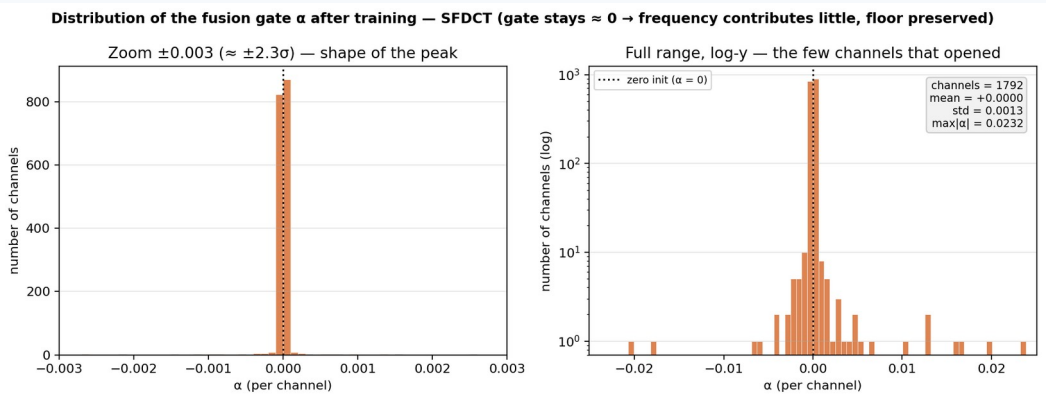
+0.0000

mean alpha

0.0232

max |alpha|

- std = **0.0013** — most channels stay near zero
- Only a **small fraction** of channels open selectively
- The gate acts as a **direct meter** of frequency contribution
- Consistent with measured gain being small



Key message: The zero-initialised gate is both a safe-fusion design (model starts = backbone) and a direct meter of frequency contribution. The meter reads: gain is small, selective, and honest — max $|\alpha| = 0.0232$ across 1792 channels.

SFDCT-HFF Variant and Liveness Reuse

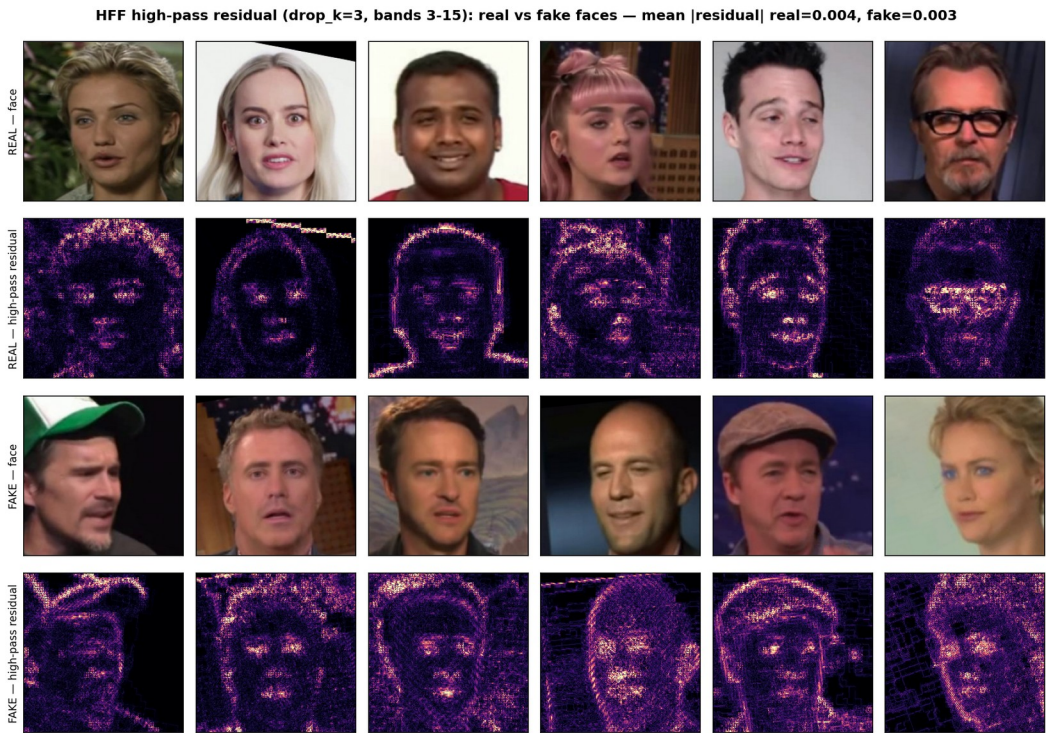
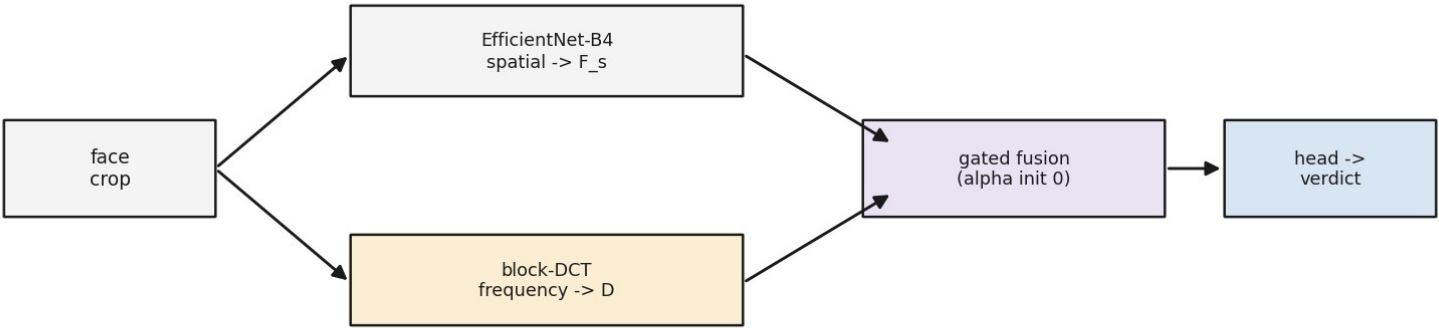
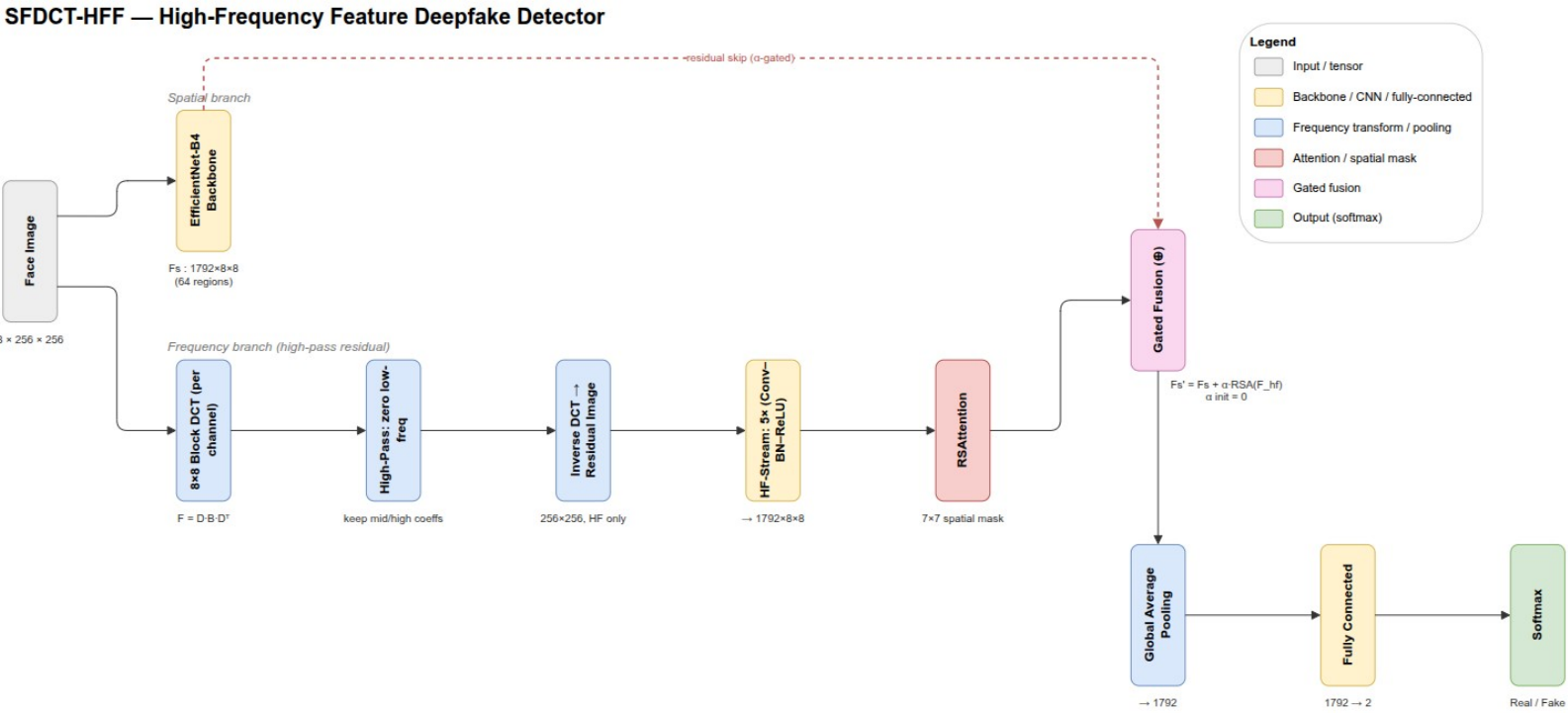
SFDCT (Baseline Variant)	SFDCT-HFF (High-Frequency Filtered)
• Frequency representation: band statistics • Keep all 16 bands (after dropping DC) • 48-dim descriptor per block • Direct statistical input to fusion gate Result: AUC = 0.7572	• Frequency representation: high-pass residual image • Zero lowest DCT bands, inverse DCT • Multi-scale Conv-BN-ReLU reads residual • Residual-guided attention steers spatial branch Result: AUC = 0.7695

Controlled comparison: Same B4 backbone, same zero-init gate, same loss. Only the frequency carrier changes — any gap is attributable to representation choice.

Liveness Module (Secondary)

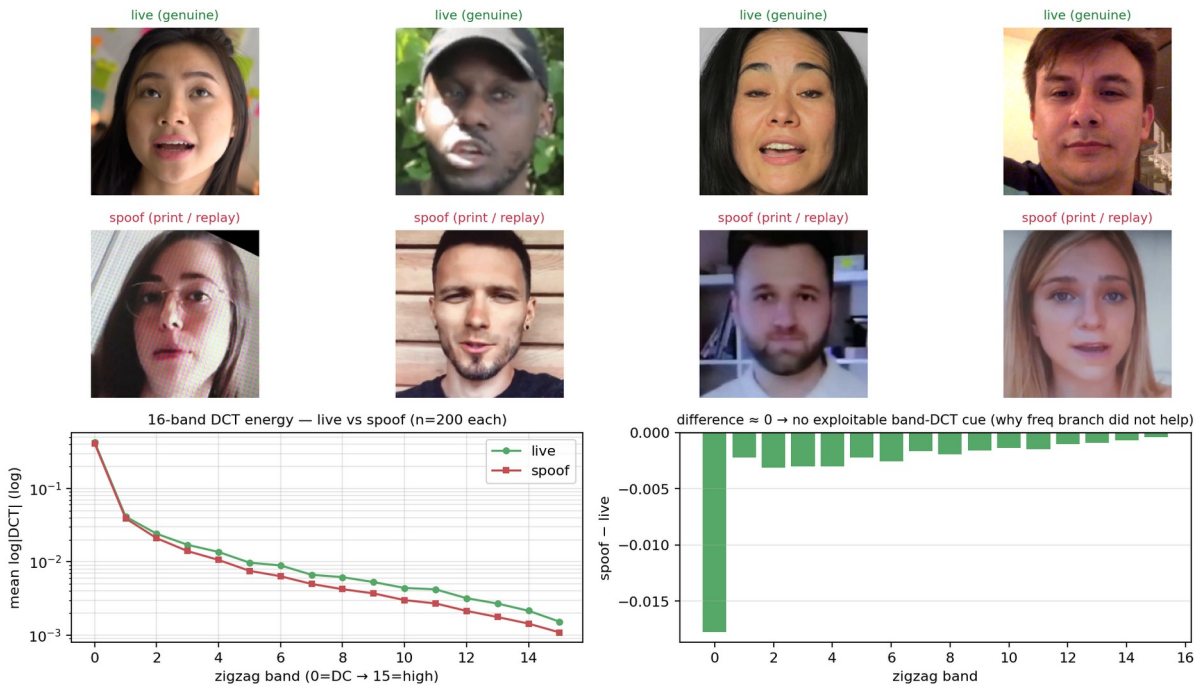
- **Reuses the same design:** B4 baseline vs. B4 + block-DCT, no new frequency design
- **Physical basis:** Screen replay leaves re-capture peaks; print leaves dot patterns — both visible in frequency domain
- **Cascade role:** Liveness runs first as a pre-filter; a spoof or uncertain verdict stops the request before deepfake computation

Key message: SFDCT-HFF changes only the frequency carrier (residual image vs. band stats), making it a controlled experiment. The same design is reused for liveness as a pre-filter cascade.

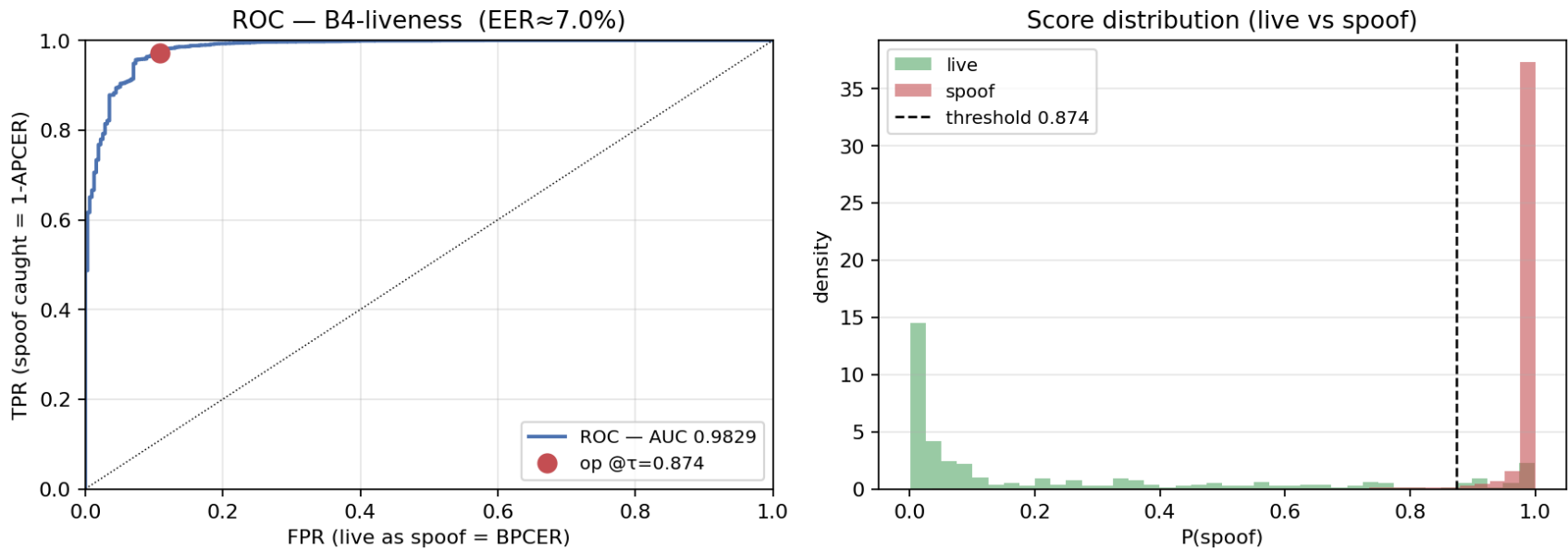


Liveness Detection (Secondary Module)

Liveness on LCC-FASD: spoof differs from live SPATIALLY but barely in the per-band DCT spectrum
→ the spatial B4 head (AUC 0.9829) is not improved by the frequency branch (B4+DCT 0.9776)



B4-liveness on LCC-FASD evaluation (n=7580, 314 live / 7266 spoof)



03

CHAPTER THREE

Results

Cross-Dataset Protocol and Honest Reading

3.1 Cross-Dataset Protocol

3.2 Cross-Dataset Results and Honest Reading

Cross-Dataset Protocol: The Real Test

Property	Training: FF++	Testing: Celeb-DF-v2
Real videos	1,000	590
Fake videos	4 methods (from 1K real)	5,639
Frames/video	32 (evenly sampled)	32 (evenly sampled)

Protocol Design

- **Train only on FF++**, test only on Celeb-DF-v2
- Test set **never seen** during training
- 32 frames/video, **fixed** for all configs
- Metric: **frame-level AUC** (threshold-free)
- Both datasets skew fake (~1:4 real:fake)



Sanity check: Baseline EfficientNet-B4 (spatial only) achieves AUC = **0.7497**, matching the published value of **0.7487** (diff ~0.001). This confirms the evaluation pipeline is correctly built.

Why Cross-Dataset Matters for eKYC

- **Realistic threat model:** Attackers in eKYC always use **newer forgery tools** than those in the training set — the detector is frozen at training time
- **In-dataset evaluation is optimistic:** Random train/test split on the same dataset measures memorisation, not generalisation
- **Cross-dataset = generalisation test:** Different forgery methods, different face distributions, different compression pipelines

Key message: The baseline reproduces the published number (0.7497 vs 0.7487), confirming the pipeline. Every subsequent score change can be attributed to model design, not setup error.

Cross-Dataset Results and an Honest Reading

Model	Frame AUC	vs B4	Floor?
B4 (spatial only)	0.7497	—	Yes
SFDCT (band stats)	0.7572	+0.0075	Yes
SFDCT-HFF (full)	0.7695	+0.0198	Yes
<i>Published phase-based freq.</i>	<i>0.7650</i>	<i>+0.0153</i>	—

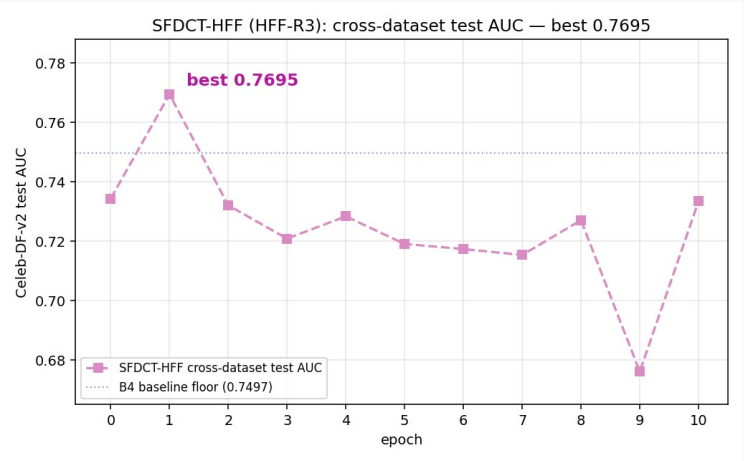
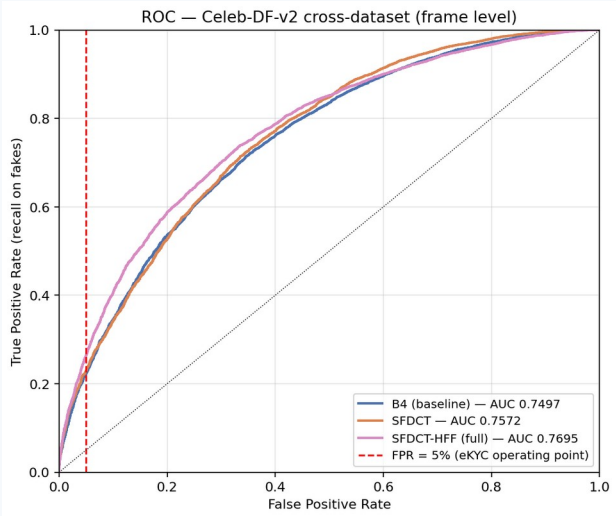
An Honest Reading

- Order **follows hypothesis**: B4 < SFDCT < HFF
- **Floor holds** at frame level for every variant
- Mean-over-run << best: best-on-test is optimistic
- Video bootstrap (518 clips): **every paired CI contains zero**
- Phase-based freq. method (0.7650) > naive SFDCT
- **No SOTA claim** — gains sit inside single-seed noise

What the Numbers Mean

At 5% false-positive rate (eKYC requirement), the model catches ~23% of deepfakes. This positions SFDCT-HFF as a **risk-screening layer with explanation** (Grad-CAM heatmap), not an absolute gate. The output is a signal for human review, not a final verdict.

Key message: The order follows the hypothesis and the floor holds, but at one random seed every paired confidence interval contains zero. This is an honest small improvement, not a superiority claim. A phase-based frequency method scores higher (0.7650).



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CHAPTER FOUR

Conclusion

Contributions, Limitations & Future Directions

4.1 Conclusion and Contributions

4.2 Limitations and Development Directions

Conclusion and Contributions

CONTRIBUTION 01

Zero-Init Floor Guarantee

The gated fusion gate starts at $\alpha=0$, so the model begins exactly equal to the proven backbone. Attaching the frequency branch can never hurt at initialisation.

CONTRIBUTION 02

Honest Evaluation with Bootstrap CI

Baseline reproduces published number. Every paired bootstrap CI contains zero at one seed — reported transparently. No superiority claim is made.

CONTRIBUTION 03

End-to-End Explainable System on CPU

Full pipeline runs on CPU (~1s/image). Returns fake probability, verdict, Grad-CAM heatmap, and DCT spectrum for human reviewer transparency.

Summary Numbers

0.7497

B4 baseline AUC

0.7572

SFDCT AUC

0.7695

SFDCT-HFF AUC

~0.98

Liveness AUC

~1s

CPU latency / image

Key message: The thesis delivers its goal — a cross-dataset deepfake detector for banking eKYC with a zero-init floor guarantee, honest evaluation, and an explainable CPU-ready system. The frequency gain is small and transparently reported, not claimed as SOTA.



Limitations and Development Directions

Limitations

1. Single run per config

One random seed; no statistical significance; every video-level CI contains zero

2. Modest gains, not SOTA

Improvements are small and within single-seed noise

3. Train/serving crop mismatch

Face alignment differs between training and deployment

4. Robustness untested

No systematic evaluation of re-compression, noise, resolution, adversarial attacks

5. Liveness: one dataset only

LCC-FASD only; cross-dataset behaviour unknown

6. No Vietnamese-face data yet

Test-only probe set in progress; no numbers reported prematurely

Development Directions

1. Multi-seed paired tests

Run multiple seeds with paired statistical tests to confirm frequency benefit

2. Fuse trained liveness score

Replace interim challenge check with the trained liveness scorer in cascade

3. Self-blended training

Integrate self-blended augmentation — strongest cross-dataset technique known

4. Vietnamese test set

Build Vietnamese-face test-only set; expand to diffusion-model forgeries

5. Harden the system

Unify train/serving crop, robustness evaluation, quantisation + pruning + distillation for edge deployment

6. Complete database design

Finish the application database schema and audit logging

Key message: Every limitation is stated transparently, and each development direction directly addresses one limitation. The priority is multi-seed validation, liveness integration, and self-blended training combined with the frequency branch.

Thank You

THANK YOU FOR YOUR ATTENTION

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Q & A

Questions and Discussion